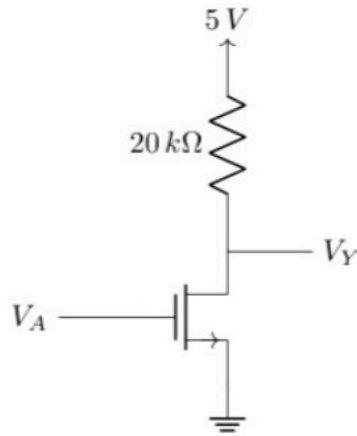


Problem 1:

For the given NMOS inverter, $V_{TN} = 0.5\text{ V}$, $K_n = 0.3 \frac{\text{mA}}{\text{V}^2}$. Find the output voltage V_Y for $V_A = 1.5$, & 5 V .

Solution:

$V_A = 5\text{ V}$: Assuming triode mode,

$$I_{DS} = I_{RD} = \frac{5 - V_Y}{20}$$

$$\Rightarrow K_n[2(V_{GS} - V_{TN})V_{DS} - V_{DS}^2] = \frac{5 - V_Y}{20}$$

$$\Rightarrow 0.5[2(5 - 0.5)V_Y - V_Y^2] = \frac{5 - V_Y}{20}$$

$$\Rightarrow V_Y = 9.07\text{ V}, 0.092\text{ V}$$

Now, V_Y cannot be 9.07 V , as that would mean I_{RD} is negative and flows in the opposite of assumed direction (i.e. from source to drain). So, $V_Y = 0.092\text{ V}$.

Verification: As $V_Y = V_{DS} = 0.092 < 4.5$, which is the overdrive voltage ($V_{GS} - V_{TN} = 5 - 0.5 = 4.5\text{ V}$), our assumption of triode mode was correct.

$V_A = 1.5\text{ V}$: Assuming saturation mode,

$$I_{DS} = I_{RD}$$

$$\Rightarrow K_n(V_{GS} - V_{TN})^2 = \frac{5 - V_Y}{20}$$

$$\Rightarrow 0.5 \times (1.5 - 0.5)^2 = \frac{5 - V_Y}{20}$$

$$\Rightarrow V_Y = -1\text{ V}$$

Verification: As $V_Y = V_{DS} = -1 < V_{OV}$, our assumption of saturation mode was wrong. Let's again assume triode mode and repeat the analysis.

$$I_{DS} = I_{RD}$$

$$\Rightarrow K_n[2(V_{GS} - V_{TN})V_{DS} - V_{DS}^2] = \frac{5 - V_Y}{20}$$

$$\Rightarrow 0.5[2(1.5 - 0.5)V_Y - V_Y^2] = \frac{5 - V_Y}{20}$$

$$\Rightarrow V_Y = 1.67 \text{ V}, 0.5 \text{ V}$$

As 1.67 V is greater than the overdrive voltage, we discard it and take 0.5 V, as it aligns with our assumption of triode mode.

Verification: As, $V_Y = 0.5 < V_{OV}$ ($V_{OV} = 1.5 - 0.5 = 1 \text{ V}$), our assumption of triode mode was correct. So, $V_Y = 0.5 \text{ V}$.

Problem 2:

For the circuit in **Problem 1**, find the maximum drain current, transition point from triode to saturation and the maximum power dissipation.

Solution:

Maximum drain current:

$I_{DS} = I_{RD} = \frac{5 - V_Y}{20}$. So, I_{DS} will rise if the output drops. Since it is an inverter, output will drop when input rises. The highest possible input to the NOT gate is $V_{DD} = 5 \text{ V}$. So, assuming $V_A = 5 \text{ V}$, we have already found $V_Y = 0.092 \text{ V}$.

$$\therefore I_{DS_{max}} = \frac{5 - 0.092}{20} = 0.2454 \text{ mA}$$

Maximum power dissipation, $P_{max} = P_1 = (5 - 0) \times I_{RD} = 5 \times \frac{5 - 0.092}{20} = 1.227 \text{ mW}$

Transition point:

At transition point, $V_{DS} = V_Y = V_{GS} - V_{TN}$. Also, $I_{DS} = K_n(V_{GS} - V_{TN})^2$. So,

$$I_{DS} = I_{RD}$$

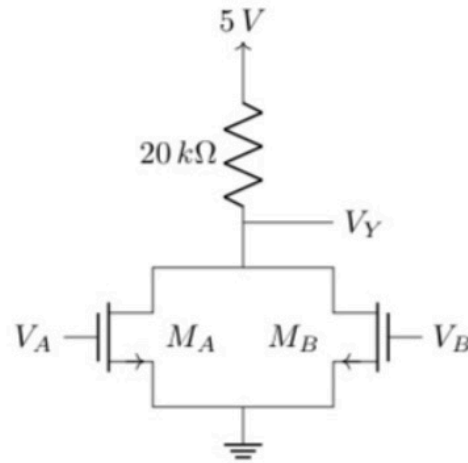
$$\Rightarrow K_n V_Y^2 = \frac{5 - V_Y}{20}$$

$$\Rightarrow V_Y = 0.66 \text{ V}, -0.76 \text{ V}$$

-0.76 V is not acceptable, since it would mean I_{DS} flows from source towards the drain. So, $V_Y = 0.66 V$.

Now, $V_A = V_{GS} = V_{DS} + V_{TN} = V_Y + V_{TN} = 0.66 + 0.5 = 1.16 V$. So, the transition point input is 1.16 V. At this input, the operating mode of the NMOS shifts from triode to saturation.

Problem 3:



For both M_A & M_B , $V_{TN} = 0.8 V$, $K_n = 0.1 \frac{mA}{V^2}$. Find the currents and output voltage for all input cases to the NOR gate.

Solution:

Case (0,0): M_A & M_B both in cut-off. So, $I_{DA} = I_{DB} = 0$. Thus, $I_{20} = I_{DA} + I_{DB} = 0$. Also, $V_Y = 5 V$.

Case (0,1) or (1,0): Let's analyze (0,1). M_A will be in cut-off. Assuming triode mode for M_B ,

$$I_{DB} = I_{20}$$

$$\Rightarrow K_n[2(V_B - V_{TN})V_Y - V_Y^2] = \frac{5 - V_Y}{20}$$

$$\Rightarrow 0.1[2(5 - 0.8)V_Y - V_Y^2] = \frac{5 - V_Y}{20}$$

$$\Rightarrow V_Y = 8.6 V, 0.29 V$$

8.6 is unacceptable as it means I_{DB} is negative. Thus, $V_Y = 0.29 V$. This aligns with our assumption of triode mode (As, $V_{OV} = 5 - 0.8 = 4.2 V$).

$$\text{Now, } I_{DB} = I_{20} = \frac{5 - 0.29}{20} = 0.235 mA$$

Case (1,1): $V_A = V_B = 5 V$. M_A & M_B both can be assumed to operate in triode mode. Thus,

$$I_{DA} + I_{DB} = I_{20}$$

$$K_n[2(V_A - V_{TN})V_Y - V_Y^2] + K_n[2(V_B - V_{TN})V_Y - V_Y^2] = \frac{5 - V_Y}{20}$$

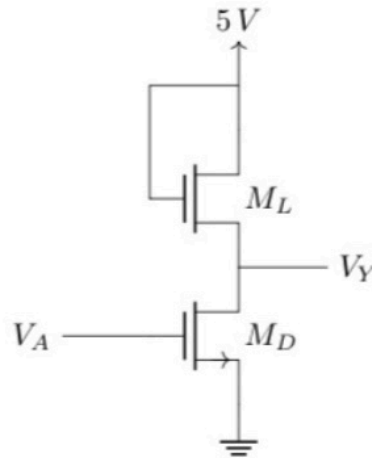
$$\Rightarrow V_Y = 4.17 \text{ V}, 0.149 \text{ V}$$

Now, 4.17 V is much closer to 4.2 V, that would drive the NMOS to saturation. So, the correct value of output voltage is 0.149V, which also aligns with the assumption of triode region operation. So, $V_Y = 0.149 \text{ V}$.

$$\therefore I_{20} = \frac{5 - 0.149}{20} = 0.243 \text{ mA}$$

$$\therefore I_{DA} = I_{DB} = \frac{0.243}{2} = 0.121 \text{ mA}$$

Problem 4:



Parameters for the driver M_D , $V_{TN_D} = 1 \text{ V}$, $K_{n_D} = 50 \frac{\mu\text{A}}{\text{V}^2}$. Parameters for the load, M_L , $V_{TN_L} = 1 \text{ V}$, $K_{n_L} = 10 \frac{\mu\text{A}}{\text{V}^2}$. Determine the output voltage and the current for both high and low input cases.

Solution:

Case (0): $V_A = 0 \text{ V}$: M_D will be in cut-off. Thus, $I_{D_D} = I_{D_L} = 0$. Now, the load M_L is an enhancement type NMOS. So, it will always stay in saturation. So, $I_{D_L} = K_{n_L}(V_{GS_L} - V_{TN_L})^2$. Since, $V_{G_L} = V_{D_L} = 5 \text{ V}$, $V_{GS_L} = V_{G_L} - V_{S_L} = 5 - V_Y$. So,

$$I_{D_L} = 10 \times 10^{-6}(5 - V_Y - 1)^2$$

$$\Rightarrow 0 = 10 \times 10^{-6}(5 - V_Y - 1)^2$$

$$\Rightarrow V_Y = 4 V$$

Case (1): $V_A = 5 V$: M_D can be assumed to be in triode. As always M_L is in saturation. So,

$$I_{D_D} = I_{D_L}$$

$$\Rightarrow K_{n_D} [2(5 - V_{TN_D})V_Y - V_Y^2] = K_{n_L} (5 - V_Y - V_{TN_L})^2$$

$$\Rightarrow 50[2(5 - 1)V_Y - V_Y^2] = 10(5 - V_Y - 1)^2$$

$$\Rightarrow V_Y = 7.65 V, 0.34 V$$

Since $7.65 V > 5 V$, the current would be negative. Thus, V_Y is $0.34 V$, which is also lower than the overdrive voltage ($5 - 1 = 4V$). Thus, our assumption is valid.

$$\therefore I_{D_D} = I_{D_L} = K_{n_L} (5 - 0.34 - 1)^2 = 0.134 mA$$

Problem 5:

Find the average power dissipation of the circuit in **Problem 4**.

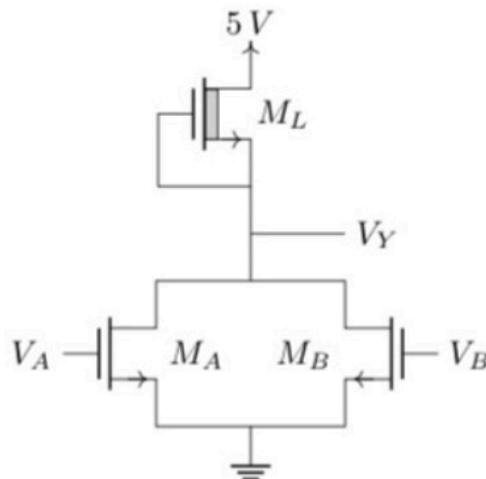
Solution:

$$\text{Case (0): } P_0 = (5 - 0) \times I_{D_L} = 5 \times 0 = 0 mW$$

$$\text{Case (1): } P_1 = (5 - 0) \times I_{D_L} = 5 \times 0.134 = 0.67 mW$$

$$P_{avg} = \frac{0 + 0.67}{2} = 0.335 mW$$

Problem 6:



It is given that $K'_n = 100 \frac{\mu A}{V^2}$, $V_{TN_A} = V_{TN_B} = 0.4 V$, $V_{TN_L} = -0.6 V$, $\left(\frac{W}{L}\right)_L = 4$, $\left(\frac{W}{L}\right)_A = \left(\frac{W}{L}\right)_B = 1$. **Note:** $K_n = K'_n \left(\frac{W}{L}\right)$. Find all the currents and the output voltage for all cases.

Solution:

Case (0,0): M_A & M_B are both off. Thus, $I_{DS_L} = I_{DS_A} = I_{DS_B} = 0$. Assuming triode mode for M_L ,

$$\begin{aligned} I_{DS_L} = 0 &= K_{n_L} [2(V_{GS_L} - V_{TN_L})V_{DS_L} - V_{DS_L}^2] \\ \Rightarrow K'_n \left(\frac{W}{L}\right)_L [2(0 - (-0.6))(5 - V_Y) - (5 - V_Y)^2] &= 0 \\ \Rightarrow 100 \times 10^{-6} \times 4 [2(0 - (-0.6))(5 - V_Y) - (5 - V_Y)^2] &= 0 \\ \Rightarrow V_Y &= 5 V, 3.8 V \end{aligned}$$

Since no current flows, there will be no drops across M_L . So, $V_Y = 5 V$ is the more acceptable answer.

Case (0,1) and (1,0): Both cases will give the same result as M_A & M_B have identical parameters. Let's analyze case (0,1). M_A will thus be off. Since, $V_{OV_L} = V_{GS_L} - V_{TN_L} = 0 - (-0.6) = 0.6 < 5$, M_L might be in saturation mode. Also, as V_Y is expected to be low in this case, we can assume saturation mode for M_B . So, assuming saturation mode for M_L and triode mode for M_B ,

$$\begin{aligned} 100 \times 10^{-6} \times 4(0 - (-0.6))^2 &= 100 \times 10^{-6} \times 1[2(5 - 0.4)V_Y - V_Y^2] \\ V_Y &= 9.04 V, 0.156 V \end{aligned}$$

9.04 V is unacceptable, as it indicates a negative drain current. So, $V_Y = 0.156$. This aligns with our assumption too.

$$\therefore I_{DS_L} = I_{DS_B} = 100 \times 10^{-6} \times 4(0 - (-0.6))^2 = 0.144 mA$$

Case (1,1): In this case, both M_A & M_B can be assumed to be in triode mode, and M_L can be assumed to be in saturation. So,

$$\begin{aligned} 100 \times 10^{-6} \times 4(0 - (-0.6))^2 &= 2 \times 100 \times 10^{-6} \times 1[2(5 - 0.4)V_Y - V_Y^2] \\ \Rightarrow V_Y &= 9.12 V, 0.078 V \end{aligned}$$

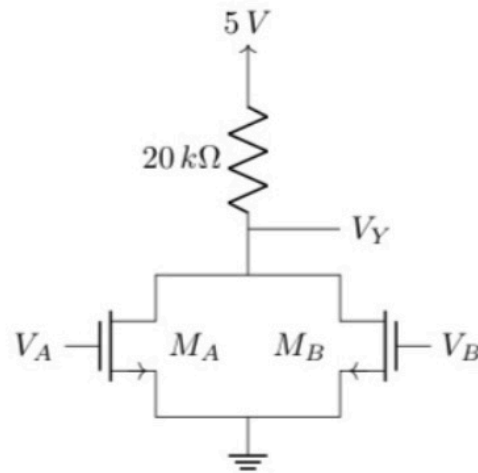
9.12 V is unacceptable due to the same reason. So, $V_Y = 0.078 V$. This also aligns with our assumption.

$$\therefore I_{DS_L} = 100 \times 10^{-6} \times 4(0 - (-0.6))^2 = 0.144 mA$$

$$\therefore I_{DS_A} = I_{DS_B} = \frac{I_{DS_L}}{2} = 0.072 mA$$

Tips: Usually saturation mode results in a greater voltage drop across the NMOS (as $V_{DS} > V_{OV}$). So, when the expected output (V_Y) is low, we may assume saturation mode for the depletion type load. Also, a low voltage at the drain of the NMOS means it is more likely to be in triode mode (as $V_{DS} < V_{OV}$). So, in the cases where V_Y is expected to be low, the driver MOSFET(s) can be assumed to be in triode mode. Although these assumptions are not 100% guaranteed to be correct, these are the most likely scenarios. So, starting the analysis with these assumptions will hopefully reduce the number of trials.

Problem 7:



For the NOR gate shown, $V_{TN_A} = 0.25\text{ V}$, $V_{TN_B} = 0.3\text{ V}$, $K_{n_A} = 0.4 \frac{\text{A}}{\text{V}^2}$, $K_{n_B} = 0.5 \frac{\text{A}}{\text{V}^2}$.

(a)	Find the lower threshold of output voltage.
(b)	Find the average power dissipation.
(c)	If a new NMOS M_C is used instead of M_B such that the drain currents of M_A & M_C are equal when both inputs are high, find the value of K_{n_C} . It is given that $V_{TN_C} = V_{TN_B} = 0.3\text{ V}$.

Solution:

(a) Case (0,1): M_A in cut-off, M_B can be assumed to be in triode. So,

$$\begin{aligned}
 I_{20} &= I_{DS_B} \\
 \Rightarrow \frac{5 - V_Y}{20} &= K_{n_B} [2(5 - V_{TN_B})V_Y - V_Y^2] \\
 \Rightarrow \frac{5 - V_Y}{20} &= 0.5 [2(5 - 0.3)V_Y - V_Y^2] \\
 \Rightarrow V_Y &= 9.45\text{ V}, 0.052\text{ V}
 \end{aligned}$$

9.45 V is greater than 5 V, so unacceptable. Thus, $V_Y = 0.052\text{ V}$. This aligns with our assumption of triode mode as $V_{OV} = 5 - 0.3 = 4.7\text{ V}$.

Case (1,0): M_A can be assumed to be in triode, M_B will be off. So,

$$\begin{aligned}
I_{20} &= I_{DS_A} \\
\Rightarrow \frac{5 - V_Y}{20} &= K_{n_A} [2(5 - V_{TN_A})V_Y - V_Y^2] \\
\Rightarrow \frac{5 - V_Y}{20} &= 0.4[2(5 - 0.25)V_Y - V_Y^2] \\
\Rightarrow V_Y &= 9.56 \text{ V}, 0.064 \text{ V}
\end{aligned}$$

For the same reasons, 9.56 is unacceptable. So, $V_Y = 0.064 \text{ V}$.

Case (1,1): Both M_A & M_B can be assumed to be in triode mode. So,

$$\begin{aligned}
I_{20} &= I_{DS_A} + I_{DS_B} \\
\Rightarrow \frac{5 - V_Y}{20} &= K_{n_A} [2(5 - V_{TN_A})V_Y - V_Y^2] + K_{n_B} [2(5 - V_{TN_B})V_Y - V_Y^2] \\
\Rightarrow \frac{5 - V_Y}{20} &= 0.4[2(5 - 0.25)V_Y - V_Y^2] + 0.5[2(5 - 0.3)V_Y - V_Y^2] \\
\Rightarrow V_Y &= 9.47 \text{ V}, 0.029 \text{ V}
\end{aligned}$$

For the same reasons, 9.47 V is unacceptable. So, $V_Y = 0.029 \text{ V}$.

$$\therefore V_{OL} = \max(0.052, 0.064, 0.029) = 0.064 \text{ V}$$

(b) For case (0,0), $I_{20} = I_{DS_A} = I_{DS_B} = 0$. So,

$$\begin{aligned}
P_{(0,1)} &= (5 - 0) \times \frac{5 - 0.052}{20} = 1.237 \text{ mW} \\
P_{(1,0)} &= (5 - 0) \times \frac{5 - 0.064}{20} = 1.234 \text{ mW} \\
P_{(1,1)} &= (5 - 0) \times \frac{5 - 0.029}{20} = 1.24275 \text{ mW} \\
P_{(0,0)} &= (5 - 0) \times 0 = 0 \text{ mW}
\end{aligned}$$

$$\therefore P_{avg} = \frac{1.237 + 1.234 + 1.24275 + 0}{4} = 0.93 \text{ mW}$$

(c) It is given that,

$$\begin{aligned}
I_{DS_A} &= I_{DS_C} \\
\Rightarrow K_{n_A} [2(5 - V_{TN_A})V_Y - V_Y^2] &= K_{n_C} [2(5 - V_{TN_C})V_Y - V_Y^2]
\end{aligned}$$

$$\Rightarrow 0.4[2(5 - 0.25)V_Y - V_Y^2] = K_{nC}[2(5 - 0.3)V_Y - V_Y^2]$$

Now,

$$I_{20} = I_{DS_A} + I_{DS_C} = I_{DS_A} + I_{DS_A} = 2I_{DS_A}$$

$$\frac{5 - V_Y}{20} = 2 \times 0.4[2(5 - 0.25)V_Y - V_Y^2]$$

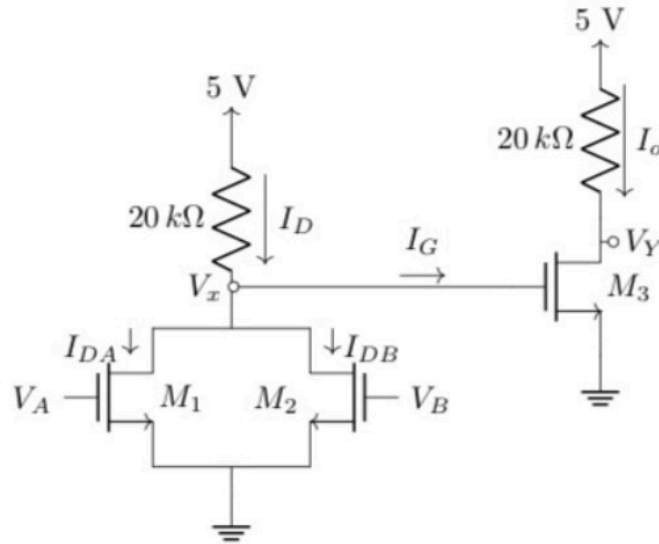
$$\Rightarrow V_Y = 9.53 \text{ V}, 0.0325 \text{ V}$$

For the same reasons, $V_Y = 0.0325 \text{ V}$. Using this value we get,

$$\Rightarrow 0.4[2(5 - 0.25)0.0325 - 0.0325^2] = K_{nC}[2(5 - 0.3)0.0325 - 0.0325^2]$$

$$\Rightarrow K_{nC} = 0.404 \frac{\text{A}}{\text{V}^2}$$

Problem 8:



Parameter	M_1	M_2	M_3
K_n	$0.3 \frac{\text{mA}}{\text{V}^2}$	$0.4 \frac{\text{mA}}{\text{V}^2}$	$0.3 \frac{\text{mA}}{\text{V}^2}$
V_{TN}	0.6 V	0.7 V	0.6 V

From the above figure, determine the currents I_D , I_{DA} , I_{DB} , I_o , V_x and V_Y for the cases- (i) both inputs are high (ii) both inputs are low. Also find the power dissipation of M_3 only for both cases.

Solution:

(i) Both inputs are high, so assume M_1 and M_2 are in triode mode.

$$I_D = I_{DA} + I_{DB}$$

$$\Rightarrow \frac{5 - V_x}{20} = 0.3[2(5 - 0.6)V_x - V_x^2] + 0.4[2(5 - 0.7)V_x - V_x^2]$$

$$\Rightarrow V_x = 8.72 \text{ V}, 0.041 \text{ V}$$

As 8.72 V is greater than 5 V, $V_x = 0.041$.

$$I_D = \frac{5 - 0.041}{20} = 0.248 \text{ mA}$$

$$I_{D_A} = 0.3[(5 - 0.6)0.041 - 0.041^2] = 0.1077 \text{ mA}$$

$$I_{D_B} = 0.4[(5 - 0.7)0.041 - 0.041^2] = 0.1404 \text{ mA}$$

Since $V_x = V_{GM_3} < 0.6 \text{ V}$, thus M_3 is in cut-off.

$$I_o = 0 \text{ mA}$$

$$I_G = 0 \text{ mA}$$

$$V_Y = 5 \text{ V}$$

(ii) Both inputs are low. So, M_1 and M_2 are in cut-off mode. Since $V_G < V_{TN}$ for both.

$$I_D = I_{D_A} = I_{D_B} = 0 \text{ mA}$$

$$V_x = 5 \text{ V}$$

$$I_G = 0 \text{ mA}$$

Since $V_x = V_{GM_3} = 5 \text{ V}$, M_3 is assumed to be in triode mode.

$$I_o = \frac{5 - V_Y}{20} = 0.3[2(5 - 0.6)V_Y - V_Y^2]$$

$$\therefore V_Y = 8.87 \text{ V}, 0.094 \text{ V}$$

For the same reasons, $V_Y = 0.094 \text{ V}$.

$$I_o = \frac{5 - 0.094}{20} = 0.2453 \text{ mA}$$

$$P_{(1,1)} = (5 - 0) \times 0.248 = 1.24 \text{ mW}$$

$$P_{(0,0)} = (5 - 0) \times 0.2453 = 1.2265 \text{ mW}$$